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Defense mechanisms and antipredator behavior in two sympatric species of spiny lobsters, *Panulirus argus* and *P. guttatus*

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Abstract The congeneric spiny lobsters *Panulirus argus* and *P. guttatus* co-occur throughout the Caribbean Sea, where they may share the coral reef habitat. Despite their phylogenetic closeness, both species have many different life-history traits that may partially explain their coexistence. However, even though both species may face the same predators, their defense mechanisms and antipredator strategies had not been compared. We compared the performance between species in 18 morphological and behavioral defense mechanisms commonly expressed by most spiny lobsters, including predator-avoidance mechanisms (activity schedule, sheltering behavior, delay to disturbance, and effect of conspecific damage-released scents on shelter choice) as well as antipredator mechanisms (body size, several parameters of the escape response and limb autospasy, clinging strength, antennal strength, and cooperative defense). As hypothesized, both species expressed all these defense mechanisms (except cooperative defense, shown only by *P. argus*), reflecting their phylogenetic closeness, but performed significantly differently in most, in accordance with their particular ontogenetic traits. Their comparative performance in individual defense mechanisms as well as the antipredator strategies displayed by groups of lobsters of each species in the presence of a common predator (the triggerfish *Balistes vetula*) showed that, in general, the defensive behavioral type of *P. argus* is more bold and that of *P. guttatus* more shy. Therefore, their distinct defensive behaviors contribute to their niche differentiation.

Introduction

The spiny lobsters *Panulirus argus* (Latreille, 1804) and *P. guttatus* (Latreille, 1804) (Decapoda: Palinuridae) co-occur throughout the Caribbean Sea and adjacent western Atlantic regions (Holthuis 1991). Morphological (George and Main 1967), ecological (George 1997), and molecular analyses (Ptacek et al. 2001) suggest that these two species are among the oldest within the genus *Panulirus* and are thus closely related phylogenetically. Therefore, in order to coexist they would require some form of niche differentiation or resource partitioning that would increase the strength of intraspecific competition relative to that of interspecific competition (Amarasekare 2003; Kneitel and Chase 2004).

Indeed, despite their similarities, both species differ in many ontogenetic traits (e.g., Briones-Fourzán 1995; Sharp et al. 1997; Lozano-Álvarez and Briones-Fourzán 2001; Acosta and Robertson 2003). In particular, *P. argus* undergoes a number of habitat shifts during its benthic life. The postlarvae of *P. argus* [~6 mm carapace length (CL)] settle on coastal shallow, vegetated areas, where the small juveniles remain for some months, showing an asocial (solitary) behavior. Once they reach 20–25 mm CL, they outgrow the protection afforded by the vegetated habitat and search for crevice-type shelters over extensive shallow areas, becoming socially gregarious (review in Butler and Herrnkind 2000). The subadults (40–45 to 75–80 mm CL) move further to adjacent coral reefs and the adults (> 80 mm CL) may occur over many kinds of habitats throughout the continental shelf to depths of ~60 m (Lozano-Álvarez et al. 1991). Individuals of *P. argus* attain large sizes and females reach sexual maturity at 75–80 mm CL (Fonseca-Larios and Briones-Fourzán 1999). The subadults and adults are highly mobile, display cooperative defense, and undergo extensive, organized migrations that may span tens or hundreds of kilometers (review in Herrnkind et al. 2001).

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By contrast, the postlarvae of *P. guttatus* (10 mm CL) settle directly on the coral reef habitat, where individuals appear to remain during their entire benthic life (Briones-Fourzán and McWilliam 1997; Sharp et al. 1997). Although the small juveniles have seldom been observed, the subadults and adults are known to be sedentary, showing a limited home range (Lozano-Álvarez et al. 2002) and a high shelter fidelity (Robertson and Butler 2003). *P. guttatus* is a small species, with females reaching sexual maturity at 30–36 mm CL (Sharp et al. 1997; Briones-Fourzán and Contreras-Ortiz 1999; Robertson and Butler 2003). Individuals of *P. guttatus* may co-den, but their aggregations appear related to reproductive activities (Briones-Fourzán and Lozano-Álvarez 2005).

The benthic distribution of *P. argus* and *P. guttatus* overlaps in the coral reef habitat. However, there are no interspecific aggressive interactions (Lozano-Álvarez and Briones-Fourzán 2001) or responses to chemical cues from live heterospecifics by either species (*P. Briones-Fourzán et al.*, unpublished data). Thus, individuals of both species may share dens. When they do, those of *P. argus* are typically positioned on the floor and those of *P. guttatus* cling to the walls or ceiling (Sharp et al. 1997; Osorio-Arciniegas 2005).

While conducting a study into the patterns of den use of both species (Lozano-Álvarez and Briones-Fourzán 2001; Osorio-Arciniegas 2005), we noticed that heterospecific individuals appeared to respond differently to the risk of predation. Because (1) predation may have a major impact on competitive interactions and coexistence of prey species (Sih 1987; Brodie et al. 1991; Seitz et al. 2001; Chase et al. 2002), (2) *P. argus* and *P. guttatus* are prey to a similar guild of predators (e.g., many species of reef fishes, Kanciruk 1980; Butler and Herrnkind 2000), and (3) promotion of coexistence is more likely if prey species show a trade-off in competitive ability versus predator defense (Chase et al. 2002), we explored the performance of each species in several defense mechanisms that may contribute to their niche differentiation.

Marine animals may use a vast array of defense mechanisms that include morphological and behavioral traits. Some defense mechanisms (“predator-avoidance mechanisms”, e.g., crypsis, immobility, temporal or spatial avoidance of predators, sheltering, burying, and slow lifestyle) operate before an attack and affect the probability of encounters with predators, whereas other defense mechanisms (“antipredator mechanisms”, e.g., armor, weaponry, escape, protean behavior, venom, distastefulness, and aggregation) operate during an attack and affect the probability of capture (Vermeij 1982; Sih 1987; Brodie et al. 1991; Hazlett et al. 2000; Herrnkind et al. 2001; Seitz et al. 2001; Barshaw et al. 2003; Lind and Cresswell 2005). It is important to distinguish between predator-avoidance and antipredator mechanisms because these two types of mechanisms often result from different selective pressures (Sih 1987; Brodie et al. 1991; Barshaw et al. 2003).

Defense mechanisms have been compared in sympatric species of decapods that belong to different families [e.g., crabs (Hazlett et al. 2000) and lobsters (Barshaw et al. 2003)] but to our knowledge, performance in defense mechanisms has not been contrasted in sympatric species of the genus *Panulirus*. We compared the performance of *P. argus* and *P. guttatus* in 18 defense mechanisms commonly expressed by lobsters, of which 4 (activity schedule, sheltering behavior, delay to first movement after disturbance, and effect of conspecific damage-released scents on shelter choice) are predator-avoidance and the remaining 14 (body size, several parameters of the escape response and limb autospasy, clinging strength, antennal strength, and cooperative defense) are antipredator mechanisms. We hypothesized that *P. guttatus* and *P. argus* could potentially express the same defense mechanisms due to their phylogenetic closeness, but that there would be interspecific differences in performance in each defense mechanism. Furthermore, because defense mechanisms may not be independent and instead interact frequently (Lind and Cresswell 2005), we also hypothesized that in a specific antipredator situation, each species would use a combination of those defense mechanisms more geared to its particular life-history and ecological requirements.

Materials and methods

We obtained information on size, activity schedule, and sheltering behavior of both species from published literature. The remaining defense mechanisms were compared through a suite of experiments conducted in the Unidad Académica Puerto Morelos del Instituto de Ciencias del Mar y Limnología, National University of Mexico, located in Puerto Morelos (20° 51'N, 86° 55'W), on the Caribbean coast of Mexico. All experiments were conducted from December 2003 to June 2004, between 11:00 and 16:00 h.

We collected lobsters from the coral reef habitat by nocturnal Scuba diving, either by hand or with hand nets (Lozano-Álvarez and Briones-Fourzán 2001; Negrete-Soto et al. 2002). Lobsters were put into a seawater container in a boat and transported to the laboratory within an hour. There, lobsters were introduced into open-flow seawater tanks, 2 m in diameter and 1 m in height, and allowed to acclimatize for at least 2 weeks. Individuals of each species were kept in separate tanks. The tanks were located in the open but under shade, and shelters for lobsters were provided in each tank. During the acclimatization period, lobsters were fed every other day with crabs or shelled mussels, but were not fed during the experiments.

Although *P. argus* attains a larger size than *P. guttatus*, most syntopic individuals of both species in the reef habitat at Puerto Morelos are 35–80 mm CL (Negrete-Soto et al. 2002; Osorio-Arciniegas 2005). Therefore, we used lobsters of both species within a similar size range

in each experiment (overall size range of *P. argus*: 33.1–77.0 mm CL; *P. guttatus*: 35.0–75.8 mm CL). Preliminary analyses disclosed no bias due to gender in either species within this size range, but we did avoid using gravid females when measuring the escape response because gravidity may affect this response in female lobsters (e.g., Cromarty et al. 1998). Also, we used only individuals in intermolt or early premolt (stages C–D₁ of Drach's scale, Lyle and MacDonald 1983) to avoid bias due to behavioral changes during the molt cycle. The behavior of lobsters in all experiments was recorded with a Sony TRV-17 mini DV videocamera. Videotapes were analyzed in a 21 in. Sony Trinitron monitor connected to a Sony SVT-L400 time-lapse VCR.

For those experiments that required the use of lobster predators, we chose the queen triggerfish (*Balistes vetula*). In predator–prey encounters, a triggerfish moves around a spiny lobster and ‘blows’ at it, trying to bite-off portions of its antennae or its eyestalks. A lobster subdued in this fashion can be quickly killed by the triggerfish (Kanciruk 1980; Lozano-Álvarez and Spanier 1997; Barshaw et al. 2003). We captured triggerfishes with hand nets in the coral reef at Puerto Morelos. The fishes were kept in a separate acclimatization tank covered with a dark mesh screen and were fed crabs twice a day.

Predator-avoidance mechanisms

Activity schedule and sheltering behavior

Being active when most predators are inactive (temporal avoidance) is a common defensive trait in animals (Sih 1987), and the way in which a species uses the habitat or the available shelters may affect its relative exposure to predators (Berry 1971). We reviewed the literature for information on the activity schedule and the sheltering behavior of *P. argus* and *P. guttatus* as recorded both in the field and in the laboratory.

Delay to first movement after disturbance

When animals are disturbed, they may move away quickly or remain stationary for some time to avoid detection (Sih 1987; Hazlett et al. 2000). We tested this behavior in 21 individuals of *P. argus* (37.5–72.8 mm CL) and 19 of *P. guttatus* (41.4–71.4 mm CL). The disturbance consisted in picking one lobster and dropping it, slightly above the surface of the water, into an opaque plastic container (60 cm in bottom diameter) with a 30 cm water depth. The lobster was allowed to sink without further disturbance. We measured the time elapsed in seconds before the lobster first moved, and videorecorded its behavior during 60 s. Delay to first movement was compared between species with a Mann–Whitney *U* test (Zar 1999) and the tapes were examined to describe and compare their behavior.

Effect of conspecific damage-released scents on shelter choice

Chemical alarm cues, including damage-released scents (the mixture of scents emanating from an injured animal) may signal the risk of danger to conspecifics. In some cases, benefits to animals receiving alarm signals extend beyond their immediate behavioral response; thus, alarm scents can mediate the learning of unknown predators or dangerous habitats (review in Chivers and Smith 1998). In the Mexican Caribbean, fishermen tail their lobster catch (*P. argus*) on board but avoid discarding the heads back into the water because they claim that this results in poor further catches in that area. No such anecdotal evidence exist for *P. guttatus*. As part of a larger study on intra- and interspecific chemical communication, we tested the effect of conspecific damage-released scents on the shelter choice of each species in Y-mazes. The full characteristics of the Y-mazes and experimental protocols are described in detail elsewhere (Briones-Fourzán and Lozano-Álvarez 2005). Briefly, a test lobster was given the choice of two identical shelters, each located in one arm of the Y. One of the shelters received water from a head-tank containing a freshly dead conspecific, cut in half lengthwise to release damage-released cues, while the other received water from a control head-tank with plain seawater. The test lobster was introduced into the start area of the Y-maze at 20:00 h and allowed to roam the Y-maze during the night. At 08:00 h the following morning, we recorded which of the two shelters it had chosen. For each species (25 *P. argus* and 28 *P. guttatus*), we analyzed the results with a one-tailed binomial test (Zar 1999). The null hypothesis was that the probability that lobsters would avoid the shelter with conspecific damage-released scents would be ≤ 0.5 .

Antipredator mechanisms

Body size

Body size may have an effect upon predation rate by affecting the outcome of predator–prey interactions: large prey gain a refuge from predation because there are fewer predators that can successfully handle them (Werner and Gilliam 1984). It is well known that *P. argus* is larger than *P. guttatus* (e.g., Negrete-Soto et al. 2002; Acosta and Robertson 2003), but we reviewed the literature for the largest sizes recorded for each species.

Escape response

Lobsters may escape from a threatening situation by tail flipping, i.e., a series of rapid abdominal flexions that results in a backward-swimming motion (Atema and Cobb 1980; Spanier et al. 1991). We analyzed 35

escape responses of *P. argus* (37.5–72.8 mm CL) and 30 of *P. guttatus* (39.0–71.4 mm CL). Experiments were run in a channel constructed with two parallel lines of stacked concrete blocks along the bottom of a tank 3 m in diameter with 30 cm water depth. The channel, 40 cm wide $\times$ 3 m long, was marked in centimeters. A wooden partition with a pulley formed a 40 $\times$ 60 cm “start compartment” at one end of the channel. To simulate an attack on the test lobster, we used a device consisting of two aluminum rods 2 m long that could be opened and closed like a scissor when handled from one end. At the other end, each rod had a perpendicular plastic rake. The rakes were introduced in the open position into the start compartment of the channel. We then introduced one lobster into the compartment. After the lobster stabilized for about 3 min and as it faced the device, the rakes were rapidly closed and the partition was raised, allowing the lobster to swim along the channel.

We analyzed seven parameters of the escape behavior (Spanier et al. 1991; Cromarty et al. 1998, 2000): (1) delay (s) to escape (how long before the lobster initiated the escape response), (2) duration of a swimming bout (s), i.e., the series of full abdominal flexions and extensions between stimulation and rest (Stentiford et al. 2000), (3) distance (m) traveled in a swimming bout, (4) overall swimming velocity (m/s), (5) acceleration (m/s^2), (6) force exerted (body weight in kg $\times$ acceleration, in newtons, N), and (7) work performed while swimming (acceleration $\times$ force, in joules, J). Body weight (BW) of experimental lobsters was estimated with the CL versus BW equations provided by Lozano-Álvarez (1992) for *P. argus* and Negrete-Soto (1988) for *P. guttatus*. Linear regressions were performed on all escape parameters against weight to double-check weight influence. Due to a non-normal distribution of the data, Mann–Whitney *U* tests were used to compare each escape parameter between species (Zar 1999).

Autospasy

When threatened by a predator many decapods, including spiny lobsters (Kanciruk 1980) may resort to autotomy (the severance of an appendage at a preformed breakage plane by means of a reflex) or to autospasy (the separation of an appendage at a preformed breakage plane when the appendage is pulled by an outside agent against resistance by the animal) (Bliss 1961). The ease with which limb severance occurs varies greatly among species (e.g., Hazlett et al. 2000; Wasson et al. 2002). To measure autospasy, we attached one end of a 40 lb fishing nickel line, 20 cm long, to a spring scale (0–3,000 g, ± 20 g) and the other end to an alligator clip. One lobster was gently held against a flat surface and, because the preformed breakage plane is located at the basi-ischium, the clip was closed over the merus of the fifth walking leg while the nickel line was slack. The spring scale was then horizontally pulled,

increasing the tension until either autospasy occurred or the scale reached its maximum tension. We measured (1) the occurrence of autospasy in each species (percent of individuals that severed the limb), (2) the delay to autospasy (s), and (3) the force (g) exerted to cause autospasy in 20 *P. argus* (42.2–71.0 mm CL) and 18 *P. guttatus* (39.6–70.8 mm CL). Occurrence of autospasy was compared between species with a Fisher exact test and the remaining parameters with Mann–Whitney *U* tests (Zar 1999). Linear regressions were calculated for each species to explore whether leg resistance was related to carapace length.

Clinging strength

Clinging to the substrate can be an effective antipredator behavior (Barshaw and Spanier 1994) and species of spiny lobsters may differ in this behavior considering their difference in robustness (Berry 1971). We measured the force needed to dislodge a spiny lobster clinging to a rough substrate outside of the water (Spanier and Lavalli 1998). A roll of wire mesh forming a compact cylinder was firmly attached to heavy construction blocks. A nickel line tie was placed around the carapace of a lobster between the third and fourth pereopods. The end of the line, which was 20 cm long, was attached to a spring scale (0–3,000 g, ± 20 g). The lobster was placed over the mesh cylinder and allowed to grip it while keeping the nickel line slack. Then, the spring scale was pulled vertically, gradually increasing the tension until the lobster was dislodged from the substratum, at which point we measured the force exerted (g) as an index of clinging strength (CS). We tested 18 *P. argus* (33.1–77.0 CL) and 17 *P. guttatus* (38.6–75.8 mm CL). Results were compared with a Mann–Whitney *U* test. The relationship between CL and CS was calculated with linear regressions. Regressions were then compared between species.

Antennal strength

Most spiny lobsters use their long and spiny antennae as offensive weapons to fend-off predators (e.g., Childress and Herrnkind 1994; Lozano-Álvarez and Spanier 1997; Herrnkind et al. 2001). However, the protective value of the antennae may depend upon their strength (i.e., their relative length and width) (Barshaw et al. 2003). To the naked eye, the antennae of *P. argus* and *P. guttatus* differ in their dimensions. Therefore, we measured the length of the antennal flagellum (LAF, cm), from the distal end of the third antennal segment to the apex of the flagellum, as well as its width (WAF = maximum diameter at the base of the flagellum, mm), in 22 *P. argus* (37.3–68.5 mm CL) and 24 *P. guttatus* (39.6–69.7 mm CL). We measured only antennal flagella that appeared to be complete, and only one per individual. We

estimated the volume of the antennal flagellum (VAF) in cm^3 as an index of its strength. We regressed each parameter against CL and compared the regressions between species.

Cooperative defense

A defense mechanism highly developed in *P. argus* is cooperative defense, in which a group of spiny lobsters protect themselves from a predator in concert with each other (Kanciruk 1980; Childress and Herrnkind 1997; Lozano-Álvarez and Spanier 1997; Herrnkind et al. 2001; Lozano-Álvarez and Briones-Fourzán 2001). Cooperative defense in *P. argus* can adopt several forms, such as a queue (a walking line of lobsters in which the spiny carapace of an individual is positioned above the abdomen of the preceding individual), a rosette (a radial formation of lobsters facing outwards with tails tucked and in contact with others in a group), or a phalanx (lobsters aligned side by side), which may be displayed both in the open and when individuals co-den (Herrnkind et al. 2001). Cooperative defense displays have not been reported in *P. guttatus* (Lozano-Álvarez and Briones-Fourzán 2001).

We conducted an experiment to measure the amount of time that groups of exposed lobsters of each species displayed cooperative defense in the presence of a predator. This experiment also allowed us to observe and describe the suite of defense mechanisms used by each species in a particular antipredator situation. One triggerfish (23–26 cm standard length) was introduced into an experimental tank 2 m in diameter and 1 m in height, devoid of shelters, and allowed to acclimatize to the new tank for 24 h without being fed. A mesh partition was then used to isolate the fish while a group of six lobsters, either *P. argus* or *P. guttatus* (i.e., a conspecific group), was introduced into the tank within a wire mesh cylinder. A few minutes later we removed the mesh cylinder to release the lobsters and then the mesh partition to release the fish. The behavior and interactions of the lobsters and the fish were videorecorded during 600 s. Four replicate tanks, each with new lobsters and fish, were used for each species. We measured the proportion of time that two or more lobsters in each tank displayed cooperative defense (i.e., queues, rosettes, or phalanxes). The proportion of time was arcsine-square root transformed to compare between species with a Student's *t* test (Zar 1999).

In addition to the two conspecific lobster experiments, we conducted an additional experiment in which each group of six lobsters consisted of three individuals of *P. argus* and the other three of *P. guttatus* (heterospecific group). We did this to examine (a) whether heterospecific individuals interacted under the risk of predation and (b) whether the proportion of time spent in cooperative defense by conspecific individuals varied in the presence of heterospecific individuals.

Results

Predator-avoidance mechanisms

Activity schedule

Individuals of both species are essentially nocturnal, but those of *P. argus* show some degree of diurnal activity. For example, they may search for mates, actively mate, and conduct their mass migrations in broad daylight (e.g., Lipcius and Herrnkind 1985; Herrnkind et al. 2001). In contrast, *P. guttatus* is exclusively nocturnal (Sharp et al. 1997). Moreover, individuals of this species conduct their activities only over a short period at night, often returning to their dens long before dawn (Lozano-Álvarez and Briones-Fourzán 2001; Segura-García et al. 2004).

Sheltering behavior

Individuals of *P. argus* use a wider variety of shelters than those of *P. guttatus* because the former may take refuge in any crevice as long as it protects their abdomen, whereas the latter prefer deep shelters into which they can fully retreat. In addition, individuals of *P. argus* tend to remain on the floor, close to the entrance of the shelter with their antennae emergent, thus remaining more visible than those of *P. guttatus*, which tend to occupy the far recesses of the den away from the entrance, often clinging to the ceiling or walls (Sharp et al. 1997; Lozano-Álvarez and Briones-Fourzán 2001; Segura-García et al. 2004; Osorio-Arciniegas 2005).

Delay to first movement after disturbance

Delay to first movement after disturbance was significantly shorter ($U = 51$; $P < 0.001$) in *P. guttatus* (median [inter-quartile range, IQ]: 0.28 [0.25–0.35] s) than in *P. argus* (0.97 [0.55–1.52] s). Most individuals of *P. guttatus* tailflipped immediately on contact with water and some swam around the container before reaching the bottom. Once on the bottom, they walked rapidly backwards with their abdomen tucked under their body even after bumping against the wall of the container, as if trying to withdraw into a crevice. By contrast, most individuals of *P. argus* sank with their legs and body extended. Once on the bottom, they either remained motionless for some time or walked slowly forwards, touching the walls of the container repeatedly with their antennae and antennules in a typical exploring behavior (Atema and Cobb 1980).

Effect of conspecific damage-released scents on shelter choice

Of the 28 *P. guttatus* tested in individual Y-maze trials, 15 (54%) chose the shelter receiving plain seawater and

13 (46%) chose the shelter receiving conspecific damage-released scents. These results did not differ significantly from random ($P=0.425$), suggesting that conspecific damage-released scents do not affect the shelter choice of *P. guttatus*. In contrast, significantly more individuals of *P. argus* (18 of 25, or 72%) chose the shelter that received plain seawater ($P=0.022$), suggesting that *P. argus* avoids shelters with conspecific damage-released cues.

Antipredator mechanisms

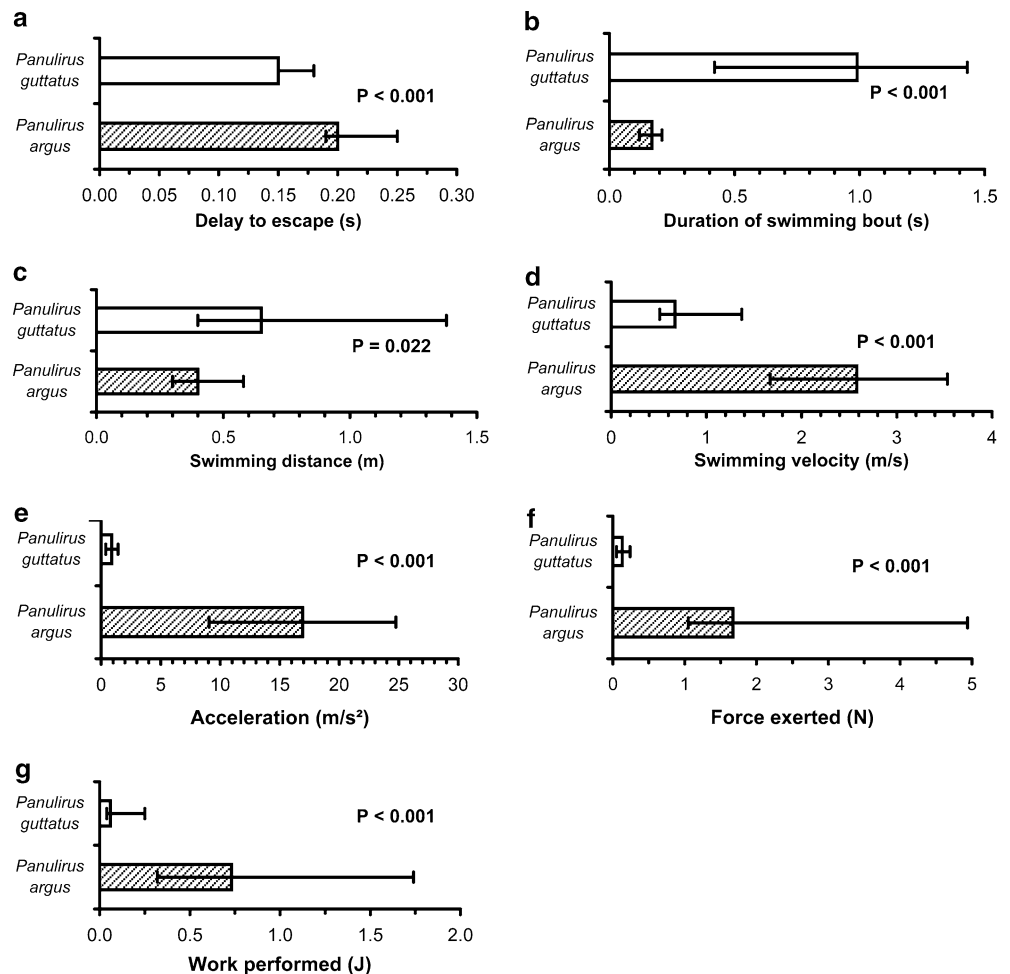
Body size

In general, animals would be expected to experience more predation when small, but they should vary in the extent of size refuge as they grow (Werner and Gilliam 1984; Hazlett et al. 2000). So far, the largest male and female of *P. argus* recorded were 204 and 180 mm CL, respectively (Lozano-Alvarez et al. 2003), whereas those of *P. guttatus* were 87.5 and 73.5 mm CL, respectively (Negrete-Soto et al. 2002; Briones-Fourzán and Contreras-Ortiz 1999).

Escape response

All seven escape parameters differed significantly between species. *P. guttatus* had a significantly shorter delay to escape than *P. argus* ($U=75$, $P<0.001$; Fig. 1a) as well as a longer duration of swimming ($U=55$, $P<0.001$; Fig. 1b) and traveled a longer distance ($U=374$, $P=0.022$; Fig. 1c). In contrast, *P. argus* had a significantly faster swimming velocity than *P. guttatus* ($U=112$, $P<0.001$; Fig. 1d) as well as a greater acceleration ($U=62$, $P<0.001$; Fig. 1e), force exerted ($U=71$, $P<0.001$; Fig. 1f), and work performed ($U=135$, $P<0.001$; Fig. 1g). In *P. guttatus*, BW was not correlated with any of the seven parameters ($r^2<0.050$; $P>0.15$ in all cases), but in *P. argus* the force exerted ($r^2=0.464$, $P<0.001$) and work performed ($r^2=0.272$, $P=0.003$) increased as BW increased. Body length (*P. guttatus*: 0.172 ± 0.018 m; *P. argus*: 0.161 ± 0.027 m; $t=1.815$, $df=62$, $P=0.08$) and weight (*P. guttatus*: 0.151 ± 0.051 kg; *P. argus*: 0.148 ± 0.071 kg; $t=0.176$, $df=62$, $P=0.86$) did not differ significantly between species, but distance per body length unit (median [IQ] *P. guttatus*: 3.69 [2.02–7.60]; *P. argus*: 2.40 [1.87–3.38]; $U=376$, $P=0.03$) and distance per BW unit (*P. guttatus*:

Fig. 1 Parameters (median $\pm$ 1 quartile) of the escape response for *Panulirus guttatus* (white bars, $N=30$) and *P. argus* (stippled bars, $N=35$). Results were compared with Mann–Whitney U tests. **a** Delay to escape: time in seconds (s) elapsed between stimulation and initiation of escape response; **b** duration of swimming bout, i.e., the series of full abdominal flexions and extensions between stimulation and rest in seconds (s); **c** distance traveled in meters (m); **d** swimming velocity (m/s); **e** acceleration (m/s^2); **f** force exerted during escape response [newtons, $\text{N} = \text{kg}/(\text{m/s}^2)$]; **g** work performed during swimming (joules, $\text{J} = \text{force} \times \text{distance}$)



4.71 [2.24–9.65] m/kg; *P. argus*: 2.66 [2.06–4.79] m/kg; $U=376$, $P=0.049$) were significantly greater in *P. guttatus*.

Autospasy

In 50% of the individuals of *P. argus* and 45% of those of *P. guttatus* either autospasy did not occur or the leg did not break at the preformed plane but at the merus. In the latter case, individuals immediately autotomized the remaining leg stump. Two individuals of *P. guttatus* showed hair-trigger autotomy, shedding their leg as the alligator clip closed over the merus. Occurrence of autospasy did not differ significantly between species (Fisher exact test, $P=0.757$). In those individuals that showed autospasy (10 *P. argus*, 9 *P. guttatus*), no significant differences occurred either in delay to autospasy (median [IQ] *P. argus*: 2.70 [1.72–3.12] s, *P. guttatus* 3.40 [2.41–4.30] s, $U=24.5$, $P=0.158$) or in leg resistance to tension (*P. argus*: 920 [740–1,300] g; *P. guttatus*: 1,000 [920–1,100] g, $U=36.5$, $P=0.724$). Leg resistance to tension was correlated with carapace length in *P. argus* ($r^2=0.442$, $P=0.036$) but not in *P. guttatus* ($r^2=0.276$, $P=0.119$).

Clinging strength

Significantly more force ($U=69.5$, $P=0.006$) was exerted to dislodge the clinging *P. guttatus* (median [IQ]: 1220 [115–1,620] g) than the *P. argus* (730 [570–1,195] g). Clinging strength (log-transformed values based on analysis of residuals) was directly related to size (CL) in both species (Fig. 2), but the slopes differed significantly ($t=3.275$; $P<0.005$). The lines intersected at a CL of 69 mm; therefore, as individuals of *P. guttatus* decreased in size below 69 mm CL, they clung to the substrate progressively more strongly than equally sized individuals of *P. argus*, whereas the situation was reversed over

69 mm CL. This CL is closer to the maximum size of *P. guttatus* than to that of *P. argus*.

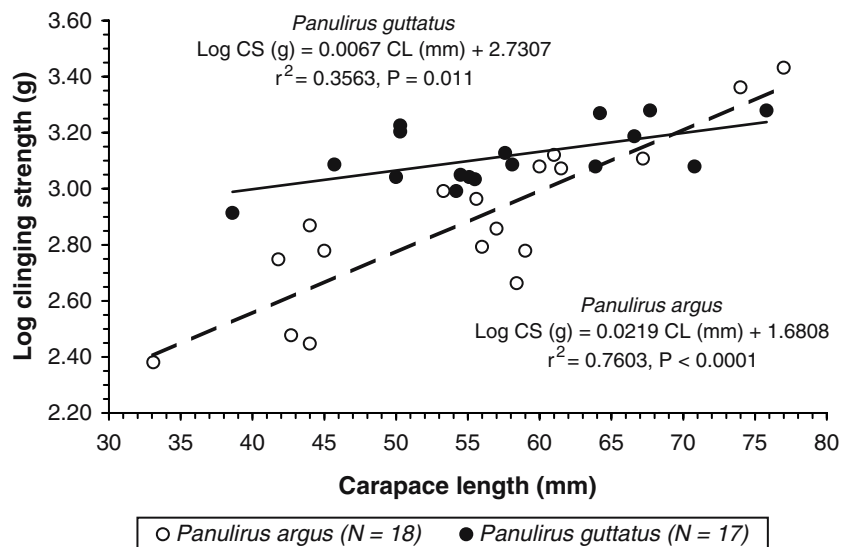
Antennal strength

In both species, antennal strength (= volume) showed a relationship with CL of the form $WAF = aCL^b$, but was greater and increased more rapidly with size in *P. argus* than in *P. guttatus* (Fig. 3a). The difference in antennal strength was due more to a difference in the width of the antennal flagellum (WAF) than in its length (LAF). Slopes of CL versus LAF regressions differed significantly between species ($t=3.307$, $P<0.005$), with the lines intersecting at 45 mm CL (Fig. 3b). Therefore, individuals of *P. guttatus* over 45 mm CL had progressively shorter antennae than individuals of *P. argus* of a similar size, while the reverse occurred for individuals below 45 mm CL. Slopes of the CL versus WAF regressions also differed significantly between species ($t=2.043$, $P<0.05$), but the lines intersected at a negative CL value (–45.3 mm CL) (Fig. 3c), showing that, in general, the antennae of *P. guttatus* are significantly more slender than those of *P. argus*.

Cooperative defense

Conspecific groups of *P. argus* exposed to the triggerfish spent most of the time displaying cooperative defense in all replicate tanks (90.2% of the time on average; range 84.5–99.8%). Their behavior was similar to that reported by Herrnkind et al. (2001). The most prevalent behaviors were queuing and rosettes. Phalanxes were seldom observed. Queues of 2–3 individuals, but more often a single queue of all six lobsters, would walk around the edge or across the bottom of the tank. When the triggerfish approached a queue, the lobsters quickly grouped and collectively pointed and waved their

Fig. 2 Relationship between carapace length (CL, mm) and log-transformed clinging strength (CS = force in grams exerted to dislodge a lobster from a substratum) in *P. guttatus* and *P. argus*. Regression lines intersected at a CL of 69 mm, indicating that as individuals of *P. guttatus* decreased in size below 69 mm CL they clung progressively more strongly than similarly sized individuals of *P. argus*, while the reverse occurred for sizes over 69 mm CL



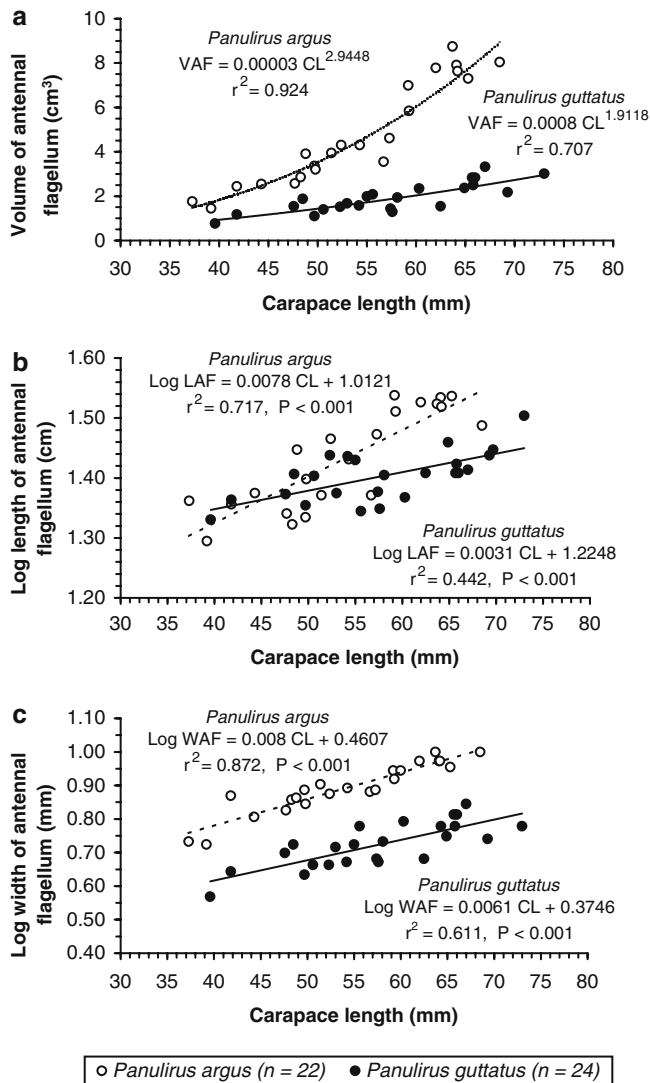


Fig. 3 *P. argus* and *P. guttatus*. Relationship between carapace length (CL, mm) and **a** volume of antennal flagella (VAF, cm³); **b** log-transformed length of antennal flagella (LAF, cm) measured from the distal end of the third antennal segment to the apex of the antennal flagella, and **c** log-transformed width of antennal flagella (WAF, mm) measured as the maximum diameter at the base of the antennal flagella. VAF is an index of the antennal strength

antennae at the fish. When the fish attacked, those lobsters closest to the fish would lunge forward, using their antennae to lash, rake and scratch the fish, which would then retreat. Tailflipping by lobsters was rare. Those individuals that tailflipped did so in a short, fast swimming bout that removed them mostly to a distance from the fish equal to or slightly longer than their antennal length. No fish managed to injure any *P. argus* and very few lobsters lost the tips of their antennae as a result of fish bites. Therefore, antennae appear to be a major weapon in *P. argus*.

In sharp contrast, there were no cooperative defense displays in conspecific groups of *P. guttatus* (0% of time in all replicate tanks). Individuals of *P. guttatus* first

dispersed over the bottom of the tank. When reaching its edge, most tended to remain wedged between the floor and the wall but some walked around the edge or tried to climb the wall of the tank. Occasionally, when two individuals walking in opposite directions met, agonistic interactions ensued (Segura-García et al. 2004). More commonly, one lobster would climb over the other and the two continued to walk their own way or remained still for a while. On several occasions, two or more *P. guttatus* clustered against the wall of the tank but we did not construe this clustering as cooperative defense because it in no way deterred the fish from attacking and the lobsters did not protect themselves in concert with each other. Only the attacked lobster reacted while nearby lobsters remained still or escaped, some by walking backwards but most by swimming all the way up to the water surface (~0.8 m) and around the tank before descending to the bottom. When an attacked *P. guttatus* pointed and waved its antennae at the fish, the fish would swim around the lobster and quickly grab an antenna with its mouth, alternately pulling the lobster away from the wall of the tank and biting-off pieces of the antenna. Individuals of *P. guttatus* did not turn around as deftly as those of *P. argus* to face the fish; consequently, most individuals of *P. guttatus* lost portions of their antennae and two were completely subdued by the fish.

In heterospecific groups, individuals of *P. argus* and *P. guttatus* did not interact and behaved similarly as when the groups were conspecific: those of *P. argus* displayed cooperative defense and those of *P. guttatus* did not. If walking heterospecifics met, each continued its own way. Agonistic interactions occurred only between individuals of *P. guttatus*. Interestingly, the fish directed most of its attacks to lone individuals or clusters of *P. guttatus*, and individuals of *P. argus* spent significantly less time (46.8% of the time on average, range 24.8–67.9%) displaying cooperative defense in heterospecific than in conspecific groups ($t = 4.352$; $df = 6$; $P = 0.005$).

Comparison of the set of defense mechanisms

Table 1 shows the comparative performances of both species in all the defense mechanisms explored. Relative to *P. guttatus*, *P. argus* had a wider activity schedule and a less cryptic sheltering behavior; a larger body size but less clinging strength; tended to avoid shelters with conspecific damage-released scents; took longer to move after disturbance and to escape; swam faster for a shorter time and distance but with greater acceleration, thus exerting more force and performing more work; had stronger antennae and displayed organized, cooperative defense. In contrast, *P. guttatus* had a narrower activity schedule and a more cryptic sheltering behavior; a smaller size but greater clinging strength; did not tend to avoid shelters with conspecific damage-released scents; reacted faster after disturbance and to a direct attack;

Table 1 Comparative relative performance of *Panulirus argus* and *P. guttatus* in 18 defense mechanisms

Defense mechanism	<i>P. argus</i>	<i>P. guttatus</i>
Activity schedule	Wider	Narrower
Sheltering behavior	Less cryptic	More cryptic
Delay to first movement after disturbance	Longer	Shorter
Effect of damage-released scents on shelter choice	Avoidance	Neutral
Body size	Larger	Smaller
Delay to escape	Longer	Shorter
Duration of swimming bout	Shorter	Longer
Swimming distance	Shorter	Longer
Swimming velocity	Faster	Slower
Swimming acceleration	Faster	Slower
Force exerted while swimming	More	Less
Work performed while swimming	More	Less
Occurrence of autospasy	Similar	Similar
Delay to autospasy	Similar	Similar
Leg resistance to tension	Similar	Similar
Clinging strength	Weaker	Stronger
Antennal strength	Stronger	Weaker
Cooperative defense	Very common	Not expressed

swam more slowly but for a longer time, covering more ground while performing less work; had weaker antennae and did not display cooperative defense. Although defense mechanisms may be tested and compared individually, they should be assessed as a package (Sih et al. 2004) because some may compensate for others (Lind and Cresswell 2005). Thus, in the presence of the triggerfish, the antipredator behavior of each species combined different mechanisms relative to the other.

Discussion

As expected, individuals of *P. argus* and *P. guttatus* were able to express all the defense mechanisms tested (except cooperative defense) but performed significantly differently in most. This was more evident when the lobsters were under the attack of the predator. Thus, over the size range of our experimental lobsters and despite intraspecific variability, *P. argus* showed a more bold and *P. guttatus* a more shy defensive behavioral type (Richardson 2001). If behavioral types are set by a combination of genotype and developmental environment (Sih et al. 2004), our results suggest that, despite their phylogenetic closeness, the different antipredator strategies of *P. argus* and *P. guttatus* result from the combinations of those defense mechanisms that would maximize their fitness according to their specific life history and ecological requirements (Sih et al. 2004; Lind and Cresswell 2005).

For example, although *P. argus* may take shelter in the coral reef, it forages over adjacent seagrass or reef flats, i.e., wide open areas characterized by ephemeral conditions, high food competition (Cox et al. 1997; Briones-Fourzán et al. 2003), and random encounters with predators. This species also migrates over open

terrain in daylight (review in Herrnkind et al. 2001). Thus, for foraging or migrating individuals of *P. argus*, a longer delay to disturbance and avoiding areas with conspecific alarm odors may help to reduce the likelihood of encounters with predators, whereas a large body size, long and thick antennae, and cooperative defense would be valuable antipredator mechanisms. When sheltering, their typical positioning on the floor of the den with their antennae exposed (Sharp et al. 1997; Herrnkind et al. 2001) might appear to facilitate their detection, but the antennae of *P. argus* are important sensory organs that respond strongly to mechanosensory stimuli such as those produced by an approaching predator (Wilkins et al. 1996). Moreover, they are also an effective weapon, more so when individuals confront a predator collectively. Thus, by lunging forward while waving their formidable antennae, co-denning individuals of *P. argus* may effectively discourage predators from entering their dens.

In contrast, *P. guttatus* has to cope throughout its whole benthic life with the high levels of predation risk that characterize the coral reef habitat to which this species is restricted. By remaining deeply sheltered during the day and conducting its activities only for short periods at night, *P. guttatus* may reduce the likelihood of encounters with predators. If an encounter occurs during a foraging excursion, the main antipredator strategy of *P. guttatus* would involve retreating quickly and as deeply as possible into any available den large enough to conceal its whole body (Lozano-Alvarez and Briones-Fourzán 2001). However, in fractal habitats such as coral reefs, the amount of potential dens tends to decrease as the size of the dens increase (Caddy 1986). Thus, for an obligated reef-dweller such as *P. guttatus*, a large body size and long and thick antennae would probably have little value. Consequently, as shown in our experiments with the triggerfish, the antennae of *P. guttatus* are not as effective a weapon as those of *P. argus*. Therefore, even if under more natural circumstances it would appear unlikely that a *P. guttatus* would confront a predator, the pointing and waving of its antennae may serve to deflect the attack to a body part that a lobster can afford to lose without dire consequences (Cooper 1998). Interestingly, the postlarvae and small juveniles of *P. guttatus* have much longer antennae than those of *P. argus* (Briones-Fourzán and McWilliam 1997), whereas our study showed that this situation is reversed at around 45 mm CL. The causes and consequences of this ontogenetic shortening of the antennae of *P. guttatus* remain undetermined because their mechanosensory capacities are unknown, as are the juvenile habits of this species.

Individuals of *P. argus* are highly sensitive to chemical stimuli (Derby 2000) and are significantly attracted to scents of live conspecifics, which can mediate gregarious behavior (Ratchford and Eggleston 1998). Our results suggest that individuals of *P. argus* also tend to avoid using shelters in the vicinity of conspecific damage-released odors, a behavior shown by other spiny

lobsters that are highly mobile and undergo ontogenetic habitat shifts [e.g., *P. cygnus* (Hancock 1974) and *P. interruptus* (Zimmer-Faust et al. 1985)]. Therefore, across the variable situations in which these highly mobile species live, it may be advantageous to avoid potentially dangerous areas signaled by conspecific damage-released cues. In contrast, the response of adult *P. guttatus* to scents of live conspecifics varies in time and appears to be related to the intensity of reproductive activity (Briones-Fourzán and Lozano-Álvarez 2005). The neutral effect of conspecific damage-released odors on the shelter choice of *P. guttatus* suggests that individuals of this sedentary species do not perceive these scents as alarm cues or, more likely, that they may offset the associated risk of danger by relying on defense mechanisms other than avoiding the area.

Our experiments to test for cooperative defense simulated a specific situation (i.e., open area with no available shelters in daylight) common to individuals of *P. argus* but not to those of *P. guttatus*. However, these experiments allowed us to compare the suite of defense mechanisms used by each species under attack by a fish, i.e., their antipredator behavior. Thus, in the absence of shelter, individuals of *P. argus* displayed a bold antipredator behavior: they readily confronted the triggerfish, turned deftly to face it at all times, showed few escape responses, engaged in cooperative defense, and relied heavily on the formidable weapon represented by their long and thick spiny antennae, effectively keeping the fish at bay. By contrast, the defensive behavior of *P. guttatus* was more shy. Their slender antennae and occasional clustering did little to deter the fish attacks; thus, they escaped in slow but protracted swimming bouts that took them up and a long distance away, effectively removing them from the visual field of the fish. In heterospecific groups, the fish appeared to rapidly assess the greater difficulty of attacking an organized group of *P. argus* as opposed to lone individuals or clusters of *P. guttatus*, and the *P. argus* also appeared to assess this situation, spending less time in cooperative defense.

The more cryptic behavior and weak antennae of *P. guttatus* may also underlie the lack of cooperative defense in this species. The clustering behavior shown by individuals of *P. guttatus* in the absence of shelter in our experimental tanks was similar to that described by Spanier and Lavalli (1998) for individuals of the slipper lobster *Scyllarides latus*, also a crevice-dweller species, held in tanks devoid of shelter. As in *S. latus* (Lavalli and Spanier 2001), clustering did not appear to confer any advantage to individuals of *P. guttatus* under attack by the triggerfish in our tanks. However, these conditions were highly artificial. No clustering behavior of *P. guttatus* has been observed in the field, probably because this species rarely leaves the reef habitat to wander in the open, if at all. In the natural habitat, if an attacked *P. guttatus* were unable to retreat into a crevice, it would more likely escape by swimming (E. Lozano-Álvarez, personal observation).

Compared to other lobsters, *P. argus* (2.96 ± 1.85 m/s) and *P. guttatus* (0.97 ± 0.72 m/s) are fast swimmers. Both swam faster than the scyllarid *S. latus* (0.32–0.47 m/s, Spanier et al. 1991), the nephropids *Nephrops norvegicus* (0.56 m/s, Stentiford et al. 2000), and *Homarus americanus* (0.56 ± 0.14 m/s, Cromarty et al. 2000); and *P. guttatus* compared favorably with the palinurid *P. interruptus* (0.90–1.20 m/s, Lindberg 1955). However, individuals of *P. argus* swam, on average, three times as fast as those of *P. guttatus*, but covering far less ground while performing about ten times as much work. If flight distance decreases with increased cost of fleeing (Ydenberg and Dill 1986; Spanier et al. 1991), this suggests greater energy expenditure during tailflipping in *P. argus* than in *P. guttatus*. Indeed, individuals of *P. argus* attacked by the triggerfish rarely tailflipped. Moreover, their fast, short swimming bouts appeared to function—rather than as a full escape response—to startle the fish while the lobsters repositioned themselves in order to regain the use of their antennae to keep the fish at bay.

In daytime surveys of sheltered lobsters across the vertical profile of the coral reef at Puerto Morelos, Osorio-Arciniegas (2005) and Lozano-Álvarez et al. (unpublished data) found that, despite some overlapping, individuals of *P. argus* tended to shelter along the lower portion of the reef, close to the bottom, whereas those of *P. guttatus* sheltered mostly over the middle and upper portions of the reef, where surge and wave action are stronger. This differential use of the reef habitat may facilitate the foraging of *P. argus* and *P. guttatus* over their respective feeding grounds, because *P. argus* forages over adjacent reef flats or seagrass beds (Cox et al. 1997; Briones-Fourzán et al. 2003) whereas *P. guttatus* forages over the coral reef habitat (Sharp et al. 1997). Individuals of *P. argus* use their legs to dig and grasp their prey, but individuals of *P. guttatus* would need to hold their position while prying-off their prey from the hard substrates over the middle and upper portions of the reef, hence its greater clinging strength. This could also underlie its tendency to cling to the walls and ceiling of dens (Sharp et al. 1997). An additional benefit of clinging strongly would be its antipredator value. For example, if a diver tries to dislodge a sheltered *P. guttatus*, it braces itself tenaciously against the walls and ceiling of the crevice, and if pulled by an appendage it may resort to autotomy (E. Lozano-Álvarez, personal observation). Other lobsters that behave similarly are *P. interruptus* (Lindberg 1955), *P. homarus*, *P. penicillatus*, *Palinurus gilchristii* (Berry 1971), and *S. latus* (Barshaw and Spanier 1994).

It might be arguable that most of our experimental *P. argus* were sexually immature whereas most *P. guttatus* were sexually mature (Briones-Fourzán and Contreras-Ortiz 1999), but we aimed to compare the performance in defense mechanisms of both species over their size-range in syntopy, i.e., when they share the coral reef habitat (Negrete-Soto et al. 2002). How

plastic their defense syndromes are over their benthic lifetime remains to be determined, especially for *P. guttatus*, because even if this species does not appear to shift habitats, the habits of its small juveniles are virtually unknown. In contrast, all the crevice-dwelling phases of *P. argus* appear to show a similar defensive behavior (see Childress and Herrnkind 1994; Herrnkind et al. 2001), whereas the small, asocial, algal-dwelling juveniles tend to rely more on wide dispersion, disruptive coloration, and cryptic habits (Butler and Herrnkind 2000).

The fact that our experiments were conducted during the day may have affected *P. guttatus* more because, unlike *P. argus*, *P. guttatus* is strictly nocturnal. However, our results concur with day and night observations on the behavior of heterospecific groups of *P. argus* and *P. guttatus* in the presence and absence of a predator (Lozano-Álvarez and Briones-Fourzán 2001), and with observations on the social behavior of *P. guttatus* across different situations (Segura-García et al. 2004; Domínguez-Gallegos 2005). On the other hand, we did not compare all the defense mechanisms potentially used by these species. Additional defense traits may include, among others, their background matching (Hazlett et al. 2000; Sherratt et al. 2005), the strength and thickness of the carapace (Barshaw et al. 2003), the amount and shape of spines over the carapace and antennae, and the use of stridulation (Bouwma and Herrnkind 2005). Also, the antipredator behavior of both lobster species might vary if attacked by predators with hunting strategies different from that of the triggerfish (Lima 2002, Lind and Cresswell 2005).

In summary, *P. argus* and *P. guttatus* express many of the common defense mechanisms used by spiny lobsters but perform differently in most. In an antipredator situation, the defense strategies of each species combine those defense mechanisms more geared to its particular life-history and ecological requirements. Moreover, as noted before, even in syntopy these sympatric congeners show little exploitative or interference competition in terms of habitat and shelter use. Thus, it appears that their resource partitioning and niche differentiation—including their antipredator behaviors—have minimized their interspecific competition, promoting their coexistence. Conducting more comparative studies in the natural environment and within a greater size range, although logistically challenging, will shed more light on the ecological and evolutionary differences and similarities between these species.

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